

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Learned to analyze technical specifications for **high-reliability systems**, translating requirements into comprehensive **test plans** and collaborating with developers on test scope.
- Executed manual and automated test cases across diverse technology stacks, identifying and documenting defects with root cause analysis to ensure system resilience for **space-grade hardware**.
- Contributed to the development and maintenance of **automated test scripts**, integrating them into **CI/CD pipelines** to ensure continuous software quality for embedded systems.
- Collaborated within an **Agile environment**, directly communicating test progress, findings, and integration status to developers and product owners for **next-generation Free Space Optics**.
- Gained practical experience with **Python** for scripting and automation tasks, supporting data pipeline development and scheduled reporting needs within the engineering team.
- Participated in production support activities for enterprise applications, including **SLA management** and **incident response** for P1/P2 issues, utilizing tools like Splunk and Dynatrace.
- Developed and containerized **Java Spring Boot** REST APIs for cross-service data exchange, implementing request validation and circuit-breaker patterns for improved system stability.

SKILLS

Embedded Systems – Embedded Software, Hardware-in-the-Loop Testing, Free Space Optics, Space-Grade Hardware

Testing & QA – Test Planning, Test Design, Test Execution, Automated Testing, Manual Testing, Defect Documentation, Root Cause Analysis, SDLC

Programming Languages – Python, Java, C, C++, JavaScript

Backend & Networking – Backend Applications, Networking Protocols

DevOps & CI/CD – CI/CD Pipelines, Jenkins, Git, Docker

Databases – SQL, PostgreSQL, MySQL, Oracle

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch

PROJECTS

Embedded Software Verification Suite

- Developed a comprehensive test suite for embedded software components of **Free Space Optics** terminals, focusing on **reliability and hardware interaction**.
- Implemented automated test scripts using **Python** to simulate various operational scenarios and validate system behavior under stress conditions, reducing manual testing effort by 40%.
- Executed hardware-in-the-loop testing, documenting and triaging defects related to **embedded system** performance and communication protocols.

Backend Application Integration Testing

- Designed and executed integration tests for **backend applications** supporting optical communication systems, ensuring seamless data flow between services.
- Created test cases to validate **networking protocols** and API interactions, identifying and resolving integration issues that could impact data transmission.
- Collaborated with developers to refine acceptance criteria and ensure that backend services met the stringent requirements for **aerospace-grade software**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)